

MANARNAK INDEX OF TROPICAL PLANTS

TAN PUAY YOK & ANGELIA SIA



A Guidebook on its Use
in the Calculation of Green Plot Ratio

LEAF AREA INDEX OF TROPICAL PLANTS

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TAN PUAY YOK & ANGELIA SIA

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Leaf Area Index of Tropical Plants

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Publication of Centre of Urban Greenery and Ecology (CUGE)

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PREFACE

A few years ago, a visitor who was being driven along the East Coast Parkway from Changi Airport to the Central Business District remarked “不简单，真的不简单!” when he saw the magnificent Rain Trees lining the expressway. Translated into English, he meant that it must be a challenging task to keep the Rain Trees in such lush condition. The expressway that has been transformed into a green tunnel by the tree canopies must have created a deep impression on him, and many visitors to Singapore alike. Reflecting on this compliment, one can ask if the admiration stemmed from an appreciation of the level of dedication put into keeping the trees in such good conditions, or if it arose from the appreciation that the management of these trees is backed by science and technical skills. We tend to think that both factors bear equal importance. No amount of science will ensure that trees will grow well unless coupled with adequate care and nurture. Conversely, care without the support of technical know-how will only result in the trees sliding into a state of decline, for trees are complex living organisms that require different management approaches at varying growth phases, and as they adapt to cope with the harsh and changing environmental conditions they are in.

Indeed, a scientific and quantitative approach towards urban greening is expected to gain importance in years to come. This is driven by the understanding that tree-planting is more than just a gardening activity. Trees are not passive elements in the landscapes. They perform multiple roles and their presence in the urban landscape often bring about positive changes in the biophysical environment that can often be counted upon to ameliorate the adverse effects of urbanization. This observation is supported by an increasing number of scientific studies that seek to quantitatively characterize such environmental contributions of trees, and the increasing use of models and tools to enable urban planners and architects to optimize the contribution of greenery in developments.

This guidebook presents the results of a study that was initiated with the same intention: to rely on science to develop a quantitative approach towards the accounting of greenery in a city or a development site. It arose from the lack of empirically determined data for a robust method to account for this quantum of greenery. The study also aimed to develop Green Plot Ratio from a concept to a greenery quantification tool. It also arose from the fact that we often forget that nature is complex, and does not conform to our assumptions or categories that we sometimes conveniently adopt for expediency and simplicity. This is perhaps also a unique demonstration that scientific concepts developed in plant eco-physiology and used primarily in production agriculture and horticulture can be adapted for use in urban greenery management.

We hope that this guidebook will be a useful resource for professionals in the landscape, horticulture and the architecture profession in Singapore.

Tan Puay Yok

Angelia Sia

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A NEED TO MEASURE “GREEN”?

“Cities will elicit their greatest loyalty, commitment, and stability when they function as places where people can confidently and consistently encounter satisfying connections with natural as well as economic and cultural wealth ...”

Stephen Kellert, co-author of The Biophilia Hypothesis

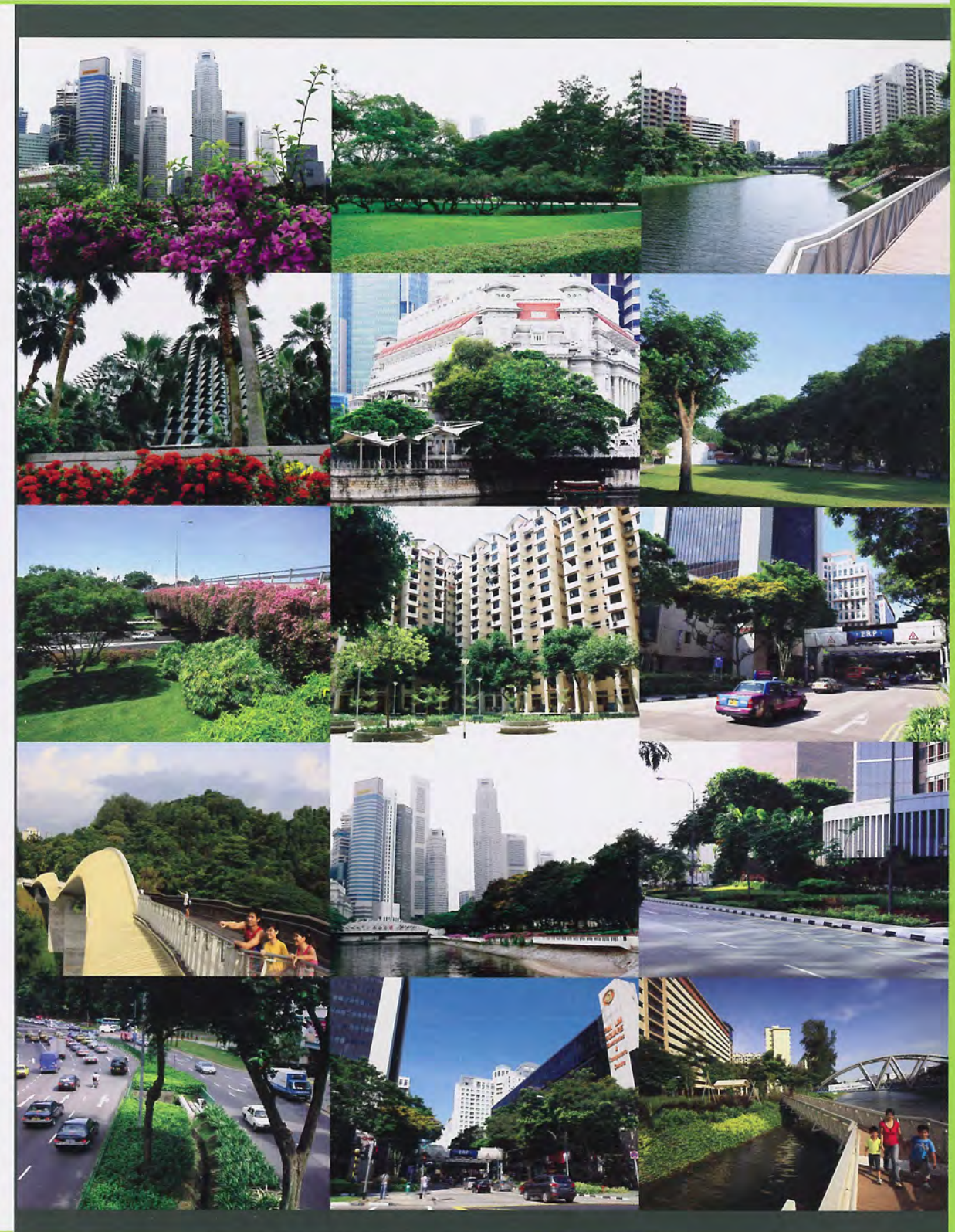
Nature and land development have often been viewed as bipolar elements within cities. This view is often reinforced by contrasting images of man-made versus natural, of humans triumphing over and taming of nature through engineering feats, and the misconception that there is no place, or need for nature in urban areas. Having one seems to infer excluding the other. However, with a growing body of knowledge in various fields on the role and importance of urban nature in cities, ranging from urban climatic to social and economic studies, there is increasing acceptance that urban development needs to be balanced with an adequate level of urban nature to ensure a high quality of life. Urban nature plays multiple roles: fostering the quality of the physical environment, catering to the physical and mental well-being of the cities’ occupants, and fostering an emotional connection with the natural heritage of cities. In this regard, to achieve a high quality of life within the cities, especially in view of the trend of increasing urbanization worldwide, urban greening and urban nature are poised to play an important role in the future development of cities.

However, what is this “balance” that needs to be struck? How does one know that balance has been achieved? Can the balance be quantitatively described? Indeed, one important consideration must be the management adage that “one cannot manage what one does not measure”.

To our knowledge, quantifying the balance between urban nature and development is a subject matter that still requires substantial research before the results can be translated into effective land-use policies and implementation guidelines. However, one could start by de-constructing nature into its components and quantitatively address each component individually. For instance, in the field of landscape ecology, there are numerous landscape metrics and indices that describe the composition and spatial configuration of green and nature areas within the city. Examples include edge density, patch density, mean patch size and connectivity. Such metrics or indices help to describe the state of the landscape and allow comparison with ideal levels or benchmarks. There are also different measures that can describe the richness in biodiversity of cities. These measures enable assessment of the current performance, and when tracked longitudinally, allow analysis of long-term trends. Examples of such measures include the Shannon biodiversity index, proximity index for species, and the Singapore Index on Cities’ Biodiversity that is being developed. These metrics and indices are also useful for comparative studies between cities.

Therefore, while a composite index that integrates various existing indices or one which adequately describes the balance between urban nature and development in a city is lacking, having specific component measures is still very useful to guide policy formulation in the management of urban nature.

In the area of urban greenery, there are also various measures that are being used to enable the active management of greenery, which in turn is needed to achieve various cost-management, environmental and ecological, as well as social objectives. A recent Singapore innovation called the Green Plot Ratio (GPR) is currently being developed. This guidebook explains how GPR can be calculated using generalized values of the key variables needed for its calculation, namely leaf area index (LAI) and canopy area. A brief description of the methodology for measurements is also provided.



HOW COULD GREENERY BE MEASURED? - AN INTRODUCTION TO GREEN PLOT RATIO

A recent innovative concept for measuring greenery in a city or a development site is the “Green Plot Ratio”. It was originally conceived by Dr Ong Boon Lay of the National University of Singapore, to be used as an “architectural and planning metric” for greenery in cities and building¹.

The distinctiveness and potential of GPR is illustrated by comparing it with other metrics that are used for the quantification of greenery. Most metrics make either one or two dimensional (number, area) measurements of greenery. The GPR, on the other hand, is three dimensional (volume) in nature. A non-exhaustive list of such metrics is described below.

Table 1. Metrics used for measurement of greenery.

CATEGORY OF METRIC	METRICS	DESCRIPTION AND CHARACTERISTICS
One- Dimensional: <i>Measurement of population numbers</i>	Inventory of Plants	The number of plants being managed, e.g. in a plantation forest <u>Characteristics:</u> - Simple to use for homogenous or heterogeneous plant populations - Does not provide information on spatial distribution
Two- Dimensional: <i>Measurement of green area</i>	Vegetation Canopy Cover	Canopy cover is the area that is under the canopy (foliage) of the vegetation, and can be measured by aerial photography or satellite imagery. <u>Characteristics:</u> - Can be used to cover large areas of interest rapidly - Additional information such as level of greenness can be derived from other measures such as Normalized Difference Vegetation Index and Enhanced Vegetation Index - Does not provide information on actual land uptake for greenery, i.e., for non- green areas that are covered by vegetation foliage - Does not account for 3-dimensional stacking of the foliage within the canopy

	Green Buffer Area	A greenery regulatory provision imposed by government agencies to safeguard peripheral planting area within private development. <u>Characteristics:</u> - Quantifies the planting area without dictating the landscape aesthetics or composition - Does not account for the three-dimensional stacking of the foliage within the canopy
	Green Area	Area of a site covered by greenery. It can be expressed as Green Area as per cent of site area, or Green Area per capita. <u>Characteristics:</u> - Does not account for the spatial distribution or composition of the green area - Does not account for the three-dimensional stacking of the foliage within the canopy
	Seattle Green Factor ² Biotope Area Factor	These are variations of a system that assigns points to various green infrastructure systems or technology. A composite score is given based on the land area occupied by the green infrastructure and specific weightings, divided by total site area. <u>Characteristics:</u> Attempts to account for various types of ecologically effective or environmentally friendly green systems
Three- Dimensional: <i>Measurement of green volume</i>	Green Plot Ratio (GPR)	Sum of total leaf area in site divided by the site area. <u>Characteristics:</u> - Accounts for vertical stacking of foliage within a canopy in the estimation of total leaf area, and hence is a approximate measure of volume (three-dimensional) rather than area. - Operationally difficult to use because of the difficulty of determining total leaf area - Expected to provide a better correlation with the environmental performance of greenery - Does not account for the landscape aesthetics or composition

GPR stands out as the only metric that quantifies the three-dimensional aspect of greenery. A more detailed explanation of GPR is given in the following section.

¹ Ong BL. 2003. Green Plot Ratio: an ecological measure for architecture and urban planning. Landscape and Urban Planning. 63: 197-211

² For more information, please see:

- Berlin Biotope Area Factor: http://www.stadtentwicklung.berlin.de/umwelt/landschaftsplanung/bff/index_en.shtml
- Seattle Green Factor: <http://www.seattle.gov/dpd/Permits/GreenFactor/Overview/>

What is Green Plot Ratio?

GPR was originally defined as the “average leaf area index (LAI) of the greenery of a site”, where the leaf area index of a plant is its one-sided leaf area per unit ground area that the plant occupies. However, it is more accurate to define Green Plot Ratio as the area weighted average leaf area index of a site, to account for unequal amount of area occupied by different plants as one would typically expect in a landscape.

In practice, in the calculation GPR, the leaf area index of each species would be multiplied by the canopy area (this yields the total leaf area in the plant) and the total for all species in the site would be divided by the site area.

GPR can therefore be also expressed as:

$$\text{GPR} = \frac{(\text{total leaf area})/(\text{site area})}{\sum (\text{LAI}_1 \times \text{Canopy Area}_1 + \text{LAI}_2 \times \text{Canopy Area}_2 + \dots + \text{LAI}_n \times \text{Canopy Area}_n)} \quad (\text{Site Area})$$

LEAF AREA INDEX

Leaf Area Index (LAI) is the one-sided area of leaf tissue per unit ground surface area.

LAI is an important parameter in horticulture production, agronomy, plant physiology, and ecology. As it directly measures the leaf area per unit ground area that a plant occupies, LAI is linked to net primary productivity of a plantation or forest, the health status of the vegetation or ecosystem, the economic yield of an orchard, modeling of plant responses to elevated CO₂, linking amount of greenery to urban climatic conditions, etc.

The application of LAI in urban greenery has thus far been very limited. This arises from both a lack of awareness of its potential as well as the practical difficulty of collecting LAI data. The Green Plot Ratio is currently the only greenery quantification metric that currently relies on LAI.

CANOPY AREA

The canopy area of a plant is simply the vertically projected area of the plant's canopy onto the ground. It is estimated as two-dimensional area of the ground area covered by the plant's canopy.

Features of Green Plot Ratio

In using the leaf area index of plants as its basis for estimation, GPR accounts for the total leaf area present in a site. As the leaves of plants are in turn responsible for many of the ecological and environmental functions of greenery, such as carbon assimilation, evaporative cooling, air pollution absorption, shading, food for herbivory insects and animals, GPR is thus a better quantification measure compared to other one or two-dimensional metrics. It enables the quantum of greenery to reflect its environmental performance more accurately.

However, the calculation of GPR is dependent on two variables, namely the leaf area index (LAI) and canopy area that are in practice, difficult to obtain. Especially for tropical ornamental plants, values of LAI and canopy area are either very limited or non-existent. LAI and canopy coverage are also not fixed values, but are expected to fluctuate with environmental conditions, health and other landscape design considerations. These practical difficulties make the calculation of GPR a challenging task and necessitate initial empirical data collection.

It is worthy to note that GPR is conceptually similar to the Gross Plot Ratio³, which is the ratio of total gross floor area to total site area. While Gross Plot Ratio is used in urban planning to control the amount of “built density”, specifically of buildings, GPR can be used to stipulate the minimum amount of “green density”. As highlighted by Ong (2003)⁴, the concept of GPR should be easily understood by the urban planning, urban design and architecture profession because of the conceptual resemblance to Gross Plot Ratio. A comparison of the two is shown in the boxed section.

GROSS PLOT RATIO	GREEN PLOT RATIO (GPR)
Ratio of Total Gross Floor Area : Site Area	Ratio of Total Leaf Area : Site Area
Stipulates “Built Density”	Stipulates “Green Density”
A standard term used in the urban planning and design profession	A newer term yet to be widely adopted
Amendments to definition of gross floor area periodically required	Developmental studies are still required
Does not regulate “look and feel” of a development site. This is supplemented by urban design guidelines	Does not regulate the aesthetics of a landscape. This is supplemented by landscape guidelines. A generic set of landscape guidelines is being developed for application in Singapore.

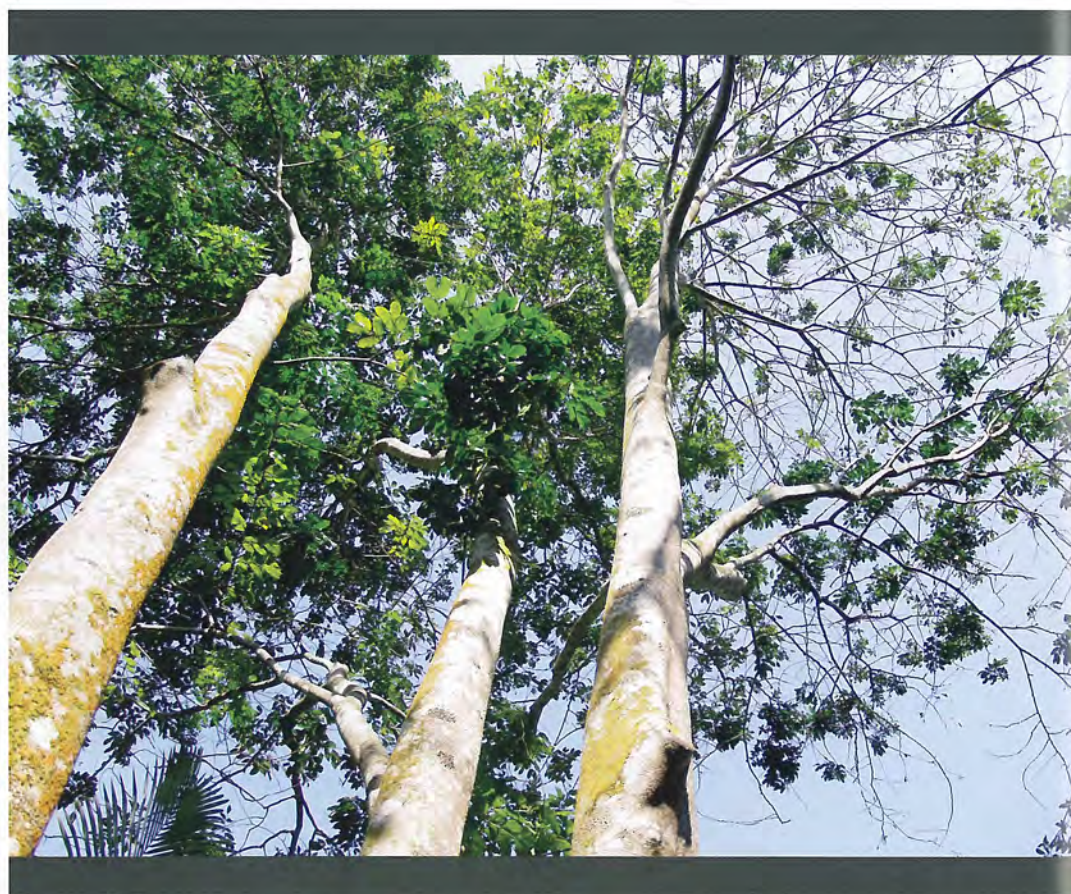
³ For more information, please see:

• Berlin Biotope Area Factor: http://www.stadtentwicklung.berlin.de/umwelt/landschaftsplanung/bff/index_en.shtml
 • Seattle Green Factor: <http://www.seattle.gov/dpd/Permits/GreenFactor/Overview/>

STUDY OF LEAF AREA INDEX AND CANOPY AREA FOR LANDSCAPE PLANTS IN SINGAPORE

For GPR of a site to be calculated, the basic information required is the leaf area index and canopy coverage of the species present in the site. While leaf area index is available for a limited range of crop and orchard plants, leaf area index values for tropical ornamental plants are practically non-existent. Yet, such numbers are critical to enable a realistic quantification of total leaf area present in a site. Adopting a set of values for different plant groups (namely trees, shrubs and ground cover) that are not empirically verified will also clearly compromise the application of GPR as a tool to relate the greenery on a site to the environmental and ecological functions of the greenery.

The Centre for Urban Greenery and Ecology (CUGE) therefore initiated a project to develop a database of the leaf area index and canopy area of common landscape plants in Singapore. This project was jointly funded by the National Parks Board and JTC Corporation. The School of Life Sciences and Chemical Technology of the Ngee Ann Polytechnic was commissioned to conduct empirical measurements on the ground. Dr Wong Suan Chin of the Australian National University provided the technical guidance on the measurement techniques. CUGE and Ngee Ann Polytechnic subsequently developed the set of guidelines for application of the leaf area index and canopy area data. A brief description of the methodology is described here.



HOW COULD LEAF AREA INDEX AND CANOPY COVERAGE BE MEASURED?

Given its significance to ecological studies, the measurement of LAI is still a science that is being developed, and one can still find new publications in the scientific literature on novel techniques being tested. A summary of measurement methods is given to mainly introduce readers to the subject matter. Techniques that have been used for measurements can be categorized into *direct*, and *indirect* measurements

Direct Measurement Methods for LAI

- Harvesting and measuring the area of all leaves in a plant within a limited land area. This is only feasible for small plants such as ground cover or turf.
- Sampling and measuring the leaf area of leaves from bigger plants. This typically involves collecting leaf litter within a specific area for deciduous trees in temperate areas. Leaf area is estimated from leaf biomass measurements and is the cumulative leaf litter over the leaf-drop period.
- Point contact method, in which a thin line or probe is passed vertically through the canopy, and LAI is determined from the number of contact points in this vertical transect.

Indirect Measurement Methods for LAI

Indirect methods rely on estimating leaf area index from the transmission of light through the canopy of the vegetation. The degree of transmission is correlated with the numbers of layers of foliage which light has to pass through, which in turn can be correlated to the leaf area index. The science behind the transmission characteristics of light through the canopy is however complex, and estimation errors can be made because of the violation of key assumptions made in the model of light transmission. Various techniques have been proposed to correct for estimation errors, which includes non-random distribution or “clumping” of leaves, as well as the temporal and spatial variation of leaf area index in the canopy. Indirect measurements are useful because of the speed and non-destructive nature of the methods.

- A variety of commercial instruments generally known as “canopy analysers” are available for indirect measurement of LAI. These include SunScan (Delta-T Devices Ltd, Cambridge, UK), LAI-2000 (Licor, Lincoln, Nebraska, USA), AccuPar (Decagon Devices, Pullman, USA), and Demon (CSIRO, Canberra, Australia).
- Before the advent of canopy analysers, LAI can also be measured by hemispherical canopy photography, which essentially uses very-wide angle lens to capture information on the amount of openings in the canopy.

For more detailed information on the theory and techniques of measuring LAI, please refer to the following references:

Breda NJJ. 2003. Ground-based measurements on leaf area index: a review of methods, instruments and current controversies. *Journal of Experimental Botany*. 54:2403-2407

Chen JM, Rich PM, Gower ST, Norman JM, Plummer S. 1997. Leaf area index of boreal forests: theory, techniques, and measurements. *Journal of Geophysical Research*. 102: 29429-29443.

Selection of Plants for the Study

A list of 300 of the most commonly used landscape plants in Singapore was generated from the National Parks Board's plant database as target plants for measurements. These are plants that have the highest populations in landscaped areas managed by NParks. The plants were categorized into the key plant groups as shown below.

Table 2. Distribution of plant groups in the field measurements of LAI and canopy area.

PLANT TYPE		APPROXIMATE DISTRIBUTION IN STUDY
TREES	Include angiosperms and gymnosperms	44%
SHRUBS	Include monocots and dicots	34%
PALMS	Include single-stemmed and multi-stemmed plants	13%
GROUND COVERS	Refers to low-growing plants used for foliage	7%
TURF	Refers to grass species	2%

Methods of Measurement of LAI

Two methods were used for measuring the leaf area index of plants in the various groups. These are indirect measurement using the plant canopy analyzer LAI-2000 (Licor, Lincoln, Nebraska, USA), and direct measurement of leaf area through partial sampling of the leaves. Indirect measurements were made using the LAI-2000 (Figure 1), which uses a fish-eye optical sensor to measure the level of light attenuation of transmission through the canopy by measuring the light level under and outside the canopy. An internal algorithm then relates the light measurements to leaf area index values. In the use of the LAI-2000, care is taken to minimize the influence of external conditions on the accuracy of measurements. In particular, measurements were made typically under overcast sky conditions where diffuse light prevail, rather than under full sun to minimize the effects of direct sunlight. Where there is non-uniform light distribution, e.g. arising from shading by adjacent large objects, view caps were used to mitigate such disturbances. A minimum of 3-4 measurements distributed around the plant's canopy per specimen was made to increase the representativeness of the measurements.



Figure 1. LAI-2000. (a) Main console. (b) Sensor in horizontal position with bubble level. (c) Range of view caps for use under different ambient conditions.

Indirect measurements using the LAI-2000 were made for trees and shrubs, as shown in Figure 2.

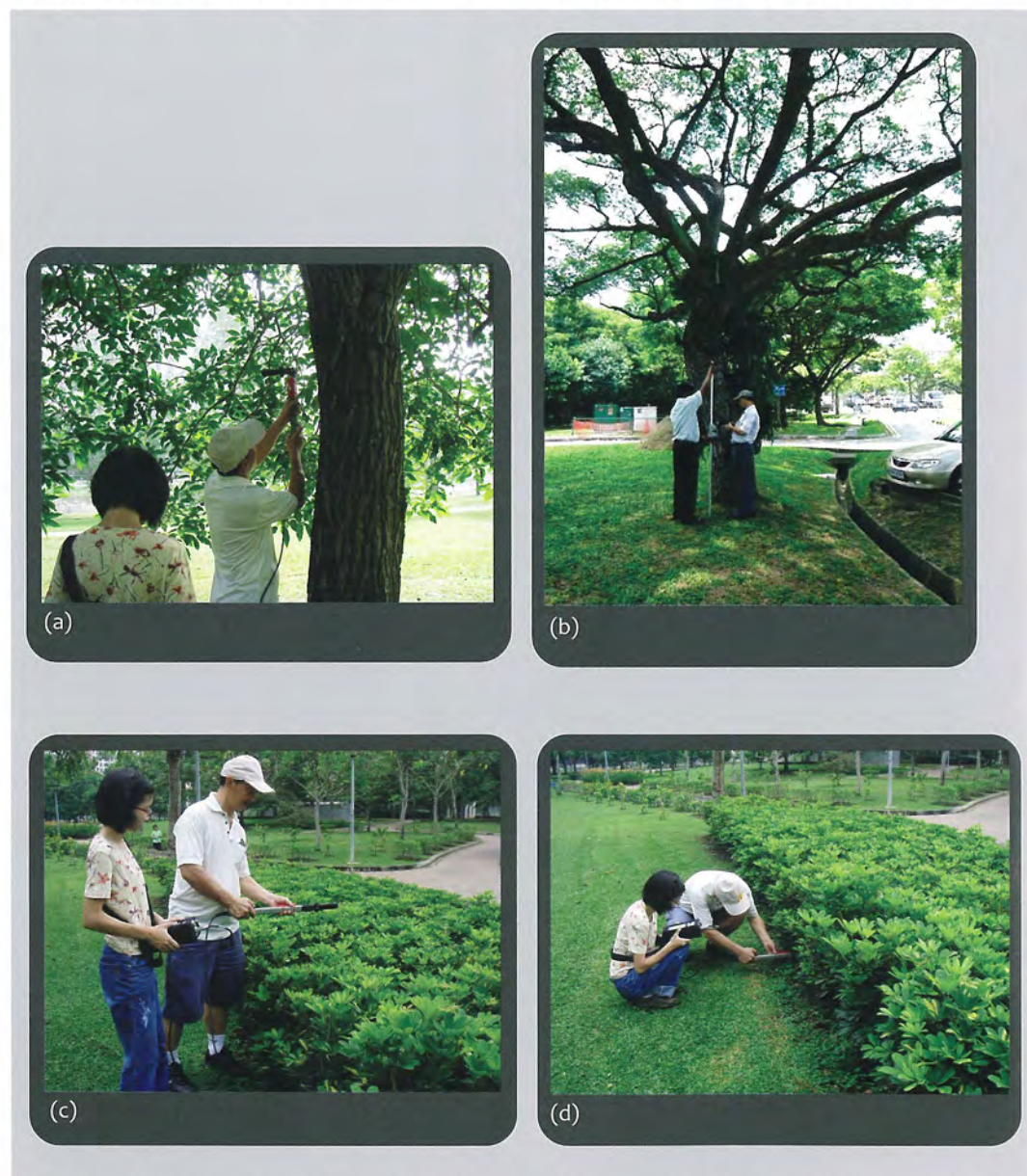


Figure 2. (a) Measurement with the LAI-2000 under the canopy of a tree. (b) Measurement with an extended pole for a taller tree. (c) Measurement of the ambient light condition above the canopy of a shrub bed. (d) Measurement of the light condition in the canopy of the shrub.

Direct measurements by partial sampling of the leaves was made for palms, ground covers and turf types. The area of sampled leaves measured with a portable leaf area meter LI-3000C connected to a conveyor belt Li-3050C (Licor, Lincoln, Nebraska, USA) (Figure 3).

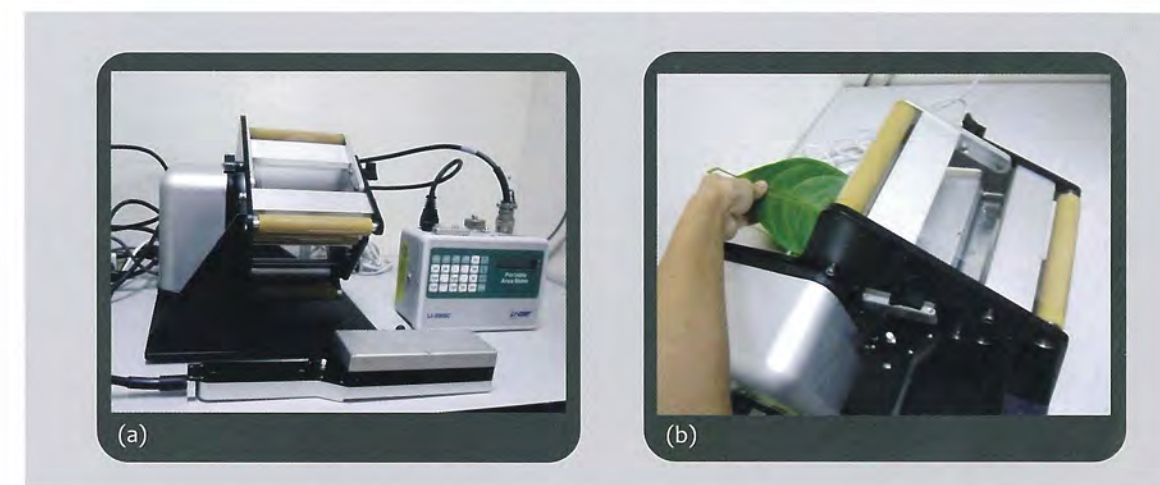


Figure 3. (a) A leaf area meter connected to a conveyor belt system for measuring leaf area. (b) A leaf being passed through the conveyor belt.

The plant selected for the measurements were sourced from a large variety of sites. These include parks and open spaces, campus within the tertiary institutions, green open spaces in public housing estates, golf courses, nurseries, and even a cemetery! For each plant species, a minimum of three specimens was measured. The standard deviation for the leaf area index measurements were within 30% of the average values for the species.

The measurement of canopy area is relatively straightforward and is not described here. However, as to be expected, the variability in canopy coverage is high, as this depends on how the plants of interest are used in the landscape, e.g., in a grove or cluster, or planted as a solitary specimen. Arising from such practical variations, application rules for the use of the LAI and canopy area need to be developed. The rules are described in the next section.

APPLICATION RULES FOR THE LEAF AREA INDEX AND CANOPY AREA

There is a great diversity of plants used in landscapes in Singapore. A conservative estimate will put the number of cultivated plants in Singapore to be between 2500 and 3000. This number will change as more plants are introduced by the nursery trade or plant collectors. Given that there are many plants not covered in the study and that the list of cultivated plants in Singapore will continue to expand, it is necessary to derive generic values that can be applied for other plants in practical fashion in the field. The following principles were used in selecting the methods to determine generic categorical values:

- To use the natural growth habit or form of plants as basis for categorizing LAI value
- The generic values should account for variations arising from plant age, and different landscaping uses of plants
- The values should be easy to use by landscaping and architecture professionals

Following the data collection on LAI and canopy area through the measurements in the field, LAI and canopy area data were arranged into different categories (e.g. by canopy shape, crown density, leaf size and shape, plant form, etc.), and statistical tests were performed to select the most robust grouping. Generic categorical values were derived based on the selected groupings using the averages of multiple plants within the categories. Table 3 summarizes the plant groups and sub categories adopted as for LAI and canopy area.

Table 3. Summary of main plant groups and the sub-categories within each group used to produce generic LAI and canopy area values.

PLANT GROUP	LAI SUB-CATEGORIES	CANOPY AREA SUB-CATEGORIES	SUB-CATEGORIES VALUES	PLANT LIST
TREES	Canopy Density, Categories (Dense, Intermediate, Open)	Girth Size	Table 4.1	Please see Appendix 1
PALMS	Solitary, Cluster	Solitary, Cluster	Table 4.2	Please see Appendix 2
SHRUBS	Dicotyledonous, Monocotyledonous	Nil - Use Planted Area	Table 4.3	Please see Appendix 3
GROUND COVERS	Nil	Nil - Use Planted Area	Table 4.4 (single group value)	NA
TURF	Nil	Nil - Use Planted Area	Table 4.5 (single group value)	NA

Application of LAI and Canopy Area Values for Trees

Trees are the most complex group to manage due to their diversity in forms and sizes. The sub-categories for LAI were derived based visual assessment of the crown density by a pool of arborists. It is important to note that statistical tests conducted showed that LAI was independent of the girth of trees, i.e., LAI did not vary significantly as the trees canopy matures. As such, only a single LAI for each canopy density class is adopted.

Table 4.1. Generic Values of LAI and Canopy Area to be used for Trees

SUB-CATEGORY	LAI VALUE TO USE	CANOPY AREA TO USE (m ²) ACCORDING TO TREE GIRTH				TREE LIST
		G1 (<0.5m)	G2 (0.5 - 1.0m)	G3 (1.0 - 1.5m)	G4 (>1.5m)	
DENSE CANOPY	4.0	12	36	80	150	Please see Appendix 1
INTERMEDIATE CANOPY	3.0	12	36	80	150	
OPEN CANOPY	2.5	12	36	80	150	

Girth Classification used by National Parks Board

- G1: Girth smaller than 0.5m.
- G2: Girth between 0.5 - 1.0m.
- G3: Girth between 1.0 - 1.5m.
- G4: Girth more than 1.5m.

Girth refers to the trunk circumference of the tree, measured at a height of 1m from the ground.

DENSE CANOPY

(close arrangement and multiple stacking of foliage within the canopy)



Mesua ferrea



Filicium decipiens



Mimusops elengi

INTERMEDIATE CANOPY

(intermediate between "dense" and "open")



Xanthostemon chrysanthus



Syzygium polyanthum



Tabebuia rosea

OPEN CANOPY

(sparse foliage arrangement or spreading canopy with little stacking of foliage within the canopy)



Plumeria rubra



Delonix regia



Samanea saman

Application of LAI and Canopy Area Values for Palms

Palms are categorized according to its key growth form, namely whether it has a single stem (or solitary palm), or has multiple stems (or cluster palm). The generic LAI and canopy area values for palms are produced according to these categories.

SINGLE-STEMMED PALM



Archontophoenix alexandrae



Bontinckia nicobarica



Bimarcia nobilis

MULTI-STEMMED PALM



Dypsis lutescens



Cyrtostachys renda



Ptychosperma macarthurii

Table 4.2. Generic Values of LAI and Canopy Area to be used for Palms

SUB-CATEGORY	LAI VALUE TO USE	CANOPY AREA TO USE (m ²)	PALM LIST
Solitary	2.5	20	Appendix 2
Cluster	4.0	17	Appendix 2

Application of LAI and Canopy Area Values for Shrubs

Flowering plants can be divided into two broad groups, the monocotyledons (or monocots) and the dicotyledons (or dicots), which are distinguished from one another by the characteristics of the developing embryo. However, in practice they can also be distinguished by the leaf pattern. Monocotyledons are characterized by parallel venation in their leaves while dicotyledons are characterized by network venation. The generic LAI values for shrubs are estimated for these two categories.

MONOCOT SHRUB

(Leaves are often long and narrow. Major leaf veins are parallel to one another)



Canna hybrids



Cordyline fruticosa
'Firebrand'



Alocasia
macrorrhiza

DICOT SHRUB

(Leaves come in different shapes and sizes. Leaf veins are usually reticulated)



Bougainvillea
'Poultoni'



Acalypha
siamensis



Ixora
'Super pink'

Table 4.3. Generic Values of LAI and Canopy Area to be used for Shrubs

SUB-CATEGORY	LAI VALUE TO USE	CANOPY AREA TO USE (m ²)	SHRUB LIST
Monocot	3.5	Use planted area of shrub bed	Appendix 3
Dicot	4.5	Use planted area of shrub bed	Appendix 3

Application of LAI and Canopy Area Values for Ground Cover

Ground covers are simply low growing plants that are used to quickly cover the areas in which they are planted. They typically require low maintenance and are used to cover areas where growing grasses is not feasible or not desired. For the determination of generic value of LAI for ground cover, the average value of the LAI of ground covers used in the field work was determined and used as the generic value.

EXAMPLE OF GROUND COVERS



Dissotis
rotundifolia



Wedelia
trilobata



Hemigraphis
alternata

Table 4.4. Generic Values of LAI and Canopy Area to be used for Ground Covers

CATEGORY	LAI VALUE TO USE	CANOPY AREA TO USE (m ²)
Ground Cover	5.0	Use planted area of ground cover

Application of LAI and Canopy Area Values for Turf

The word "turf" as used in this guidebook refers to an area covered by a matt of grass (and its roots). Common grasses used as turf in Singapore include *Axonopus compressus*, *Paspalum vaginatum*, *Zoysia matrella*, etc. To determine the generic value of LAI for turf, the average value of LAI of turf grasses measured in the field work was calculated and adopted as the generic value.

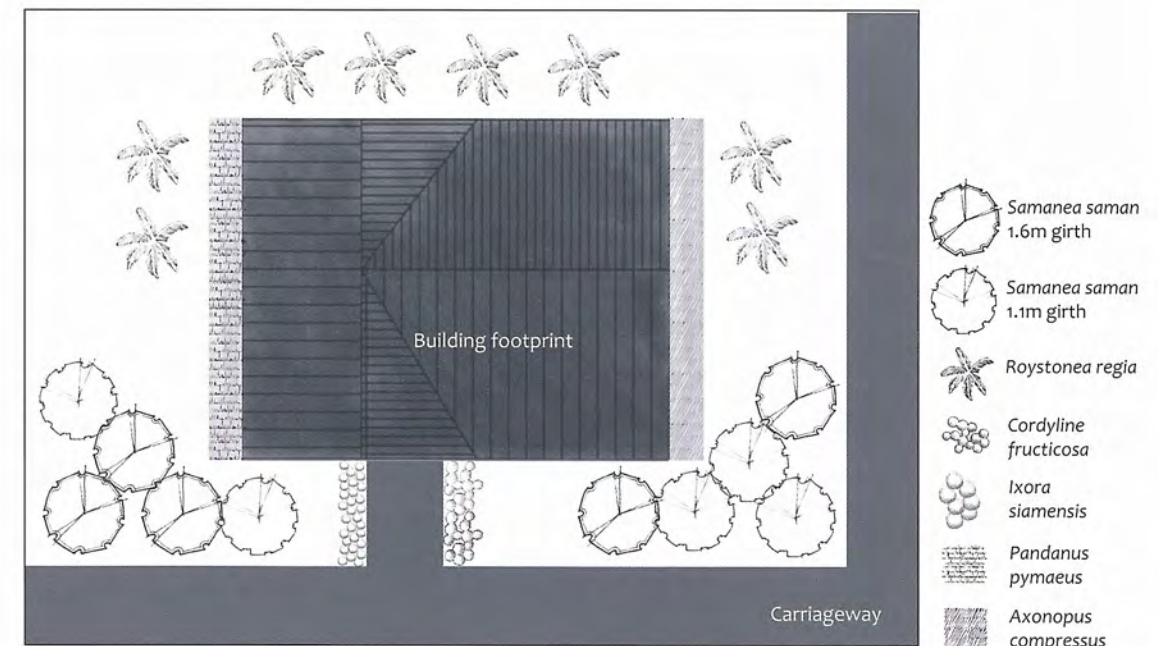
Table 4.5. Generic Values of LAI and Canopy Area to be used for Turf

CATEGORY	LAI VALUE TO USE	CANOPY AREA TO USE (m ²)
Turf	2.0	Use planted area of turf

Illustration on Use of LAI and Canopy Area Values in the Calculation of Green Plot Ratio

Suppose a landscape site has the following characteristics and plant composition:

Total Site Area	4000 sq m
Plant Composition in landscaped area	500 sq m
Trees	10 x <i>Samanea saman</i> (5 trees have girth = 1.1m; 5 trees have girth = 1.6m)
Palms	8 x <i>Roystonea regia</i>
Shrubs	200 x <i>Cordyline fruticosa</i> planted over 50 sq m
	200 x <i>Ixora siamensis</i> planted over 50 sq m
Ground cover	<i>Pandanus pyraeus</i> planted over 150 sq m
Turf	<i>Axonopus compressus</i> over 200 sq m



The calculation of total leaf area on site is illustrated below in a spreadsheet.

TREES							
SPECIES	SUB-CATEGORY	SIZE CATEGORY	LAI VALUE	CANOPY AREA (SQ M)	LEAF AREA PER PLANT (SQ M)	PLANT NUMBER	TOTAL LEAF AREA (SQ M)
(A)	(B)	(C)	(D)	(E)	(F) (F= D X E)	(G)	(H) (H= F X G)
<i>Samanea saman</i>	Open	G2	2.5	36	90	5	450
<i>Samanea saman</i>	Open	G3	2.5	80	200	5	1000

PALMS						
SPECIES	SUB-CATEGORY	LAI VALUE	CANOPY AREA (SQ M)	LEAF AREA PER PLANT (SQ M)	PLANT NUMBER	TOTAL LEAF AREA (SQ M)
(A)	(B)	(C)	(D)	(E) (E= C X D)	(F)	(G) (G= E X F)
<i>Bismarckia nobilis</i>	Solitary	2.5	20	50	8	400

SHRUBS				
SPECIES	SUB-CATEGORY	LAI VALUE (D)	PLANTED AREA (SQ M)	LEAF AREA (SQ M)
(A)	(B)	(C)	(D)	(E) (E= C X D)
<i>Cordyline fruticosa</i>	Monocot	3.5	50	175
<i>Ixora siamensis</i>	Dicot	4.5	50	225

- Obtained from
Table 4.1 - 4.5

GROUND COVER			
SPECIES	LAI VALUE	PLANTED AREA (SQ M)	LEAF AREA (SQ M)
(A)	(B)	(C)	(D) (D= B X C)
<i>Pandanus pymaeus</i>	5.0	150	750

TURF			
SPECIES	LAI VALUE	PLANTED AREA (SQ M)	LEAF AREA (SQ M)
(A)	(B)	(C)	(D) (D = B X C)
<i>Axonopus compressus</i>	2.0	200	400
TOTAL LEAF AREA			3400

Green Plot Ratio for the site = (total leaf area/ site area)
= (3400/ 4000)
= 0.85

The calculation of total leaf area on site is illustrated below in a spreadsheet.

TREES							
SPECIES	SUB-CATEGORY	SIZE CATEGORY	LAI VALUE	CANOPY AREA (SQ M)	LEAF AREA PER PLANT (SQ M)	PLANT NUMBER	TOTAL LEAF AREA (SQ M)
(A)	(B)	(C)	(D)	(E)	(F) (F= D X E)	(G)	(H) (H= F X G)
<i>Samanea saman</i>	Open	G2	2.5	36	90	5	450
<i>Samanea saman</i>	Open	G3	2.5	80	200	5	1000

PALMS						
SPECIES	SUB-CATEGORY	LAI VALUE	CANOPY AREA (SQ M)	LEAF AREA PER PLANT (SQ M)	PLANT NUMBER	TOTAL LEAF AREA (SQ M)
(A)	(B)	(C)	(D)	(E) (E= C X D)	(F)	(G) (G= E X F)
<i>Bismarckia nobilis</i>	Solitary	2.5	20	50	8	400

SHRUBS				
SPECIES	SUB-CATEGORY	LAI VALUE (D)	PLANTED AREA (SQ M)	LEAF AREA (SQ M)
(A)	(B)	(C)	(D)	(E) (E= C X D)
<i>Cordyline fruticosa</i>	Monocot	3.5	50	175
<i>Ixora siamensis</i>	Dicot	4.5	50	225

- Obtained from
Table 4.1 - 4.5

GROUND COVER			
SPECIES	LAI VALUE	PLANTED AREA (SQ M)	LEAF AREA (SQ M)
(A)	(B)	(C)	(D) (D= B X C)
<i>Pandanus pymaeus</i>	5.0	150	750

TURF			
SPECIES	LAI VALUE	PLANTED AREA (SQ M)	LEAF AREA (SQ M)
(A)	(B)	(C)	(D) (D = B X C)
<i>Axonopus compressus</i>	2.0	200	400
TOTAL LEAF AREA			3400

Green Plot Ratio for the site = (total leaf area/ site area)
= (3400/ 4000)
= 0.85

Appendix 1. List Of Trees According To Different Canopy Density Class

To identify the LAI value for trees that are not listed in the table, select the category where the same Genus can be found. Where trees of the same Genus cannot be located, the Family should be used. The Family of the species can be determined in the Plants Database accessible from this website (<http://plants.usda.gov>). In instances where the Family or Genus occur in more than one category, a conservative approach should be used to use the lower LAI category.

Trees with open canopy		
1 <i>Acacia mangium</i>	26 <i>Senna spectabilis</i>	51 <i>Tabebuia argentea</i>
2 <i>Adenanthra pavonina</i>	27 <i>Sesbania grandiflora</i>	52 <i>Tabebuia chrysantha</i>
3 <i>Artocarpus elasticus</i>	28 <i>Sesbania grandiflora</i> 'Alba'	53 <i>Taxodium distichum</i>
4 <i>Eucalyptus camaldulensis</i>	29 <i>Severinia buxifolia</i>	54 <i>Terminalia catappa</i>
5 <i>Eucalyptus deglupta</i>	30 <i>Shorea bracteolata</i>	55 <i>Terminalia ivorensis</i>
6 <i>Ficus lyrata</i>	31 <i>Shorea caudatus</i>	56 <i>Terminalia kaernbachii</i>
7 <i>Ficus pleurocarpa</i>	32 <i>Shorea macroptera</i>	57 <i>Terminalia mantaly</i>
8 <i>Juniperus chinensis</i>	33 <i>Shorea palembanica</i>	58 <i>Terminalia mantaly</i> 'Tricolor'
9 <i>Lepisanthes amoena</i>	34 <i>Shorea roxburghii</i>	59 <i>Terminalia muelleri</i>
10 <i>Melicope elleryana</i>	35 <i>Shorea sumatrana</i>	60 <i>Terminalia sericea</i>
11 <i>Mesua ferrea</i>	36 <i>Sindora echinocalyx</i>	61 <i>Tetrapleura tetraptera</i>
12 <i>Michelia alba</i>	37 <i>Sindora wallichii</i>	
13 <i>Mimusops balata</i>	38 <i>Solanum wrightii</i>	
14 <i>Moringa oleifera</i>	39 <i>Spondias dulcis</i>	
15 <i>Parmentiera cereifera</i>	40 <i>Sterculia coccinea</i>	
16 <i>Peronema canescens</i>	41 <i>Sterculia foetida</i>	
17 <i>Pinus caribaea</i>	42 <i>Sterculia macrophylla</i>	
18 <i>Plumeria pudica</i>	43 <i>Sterculia monosperma</i>	
19 <i>Pseudobombax ellipticum</i>	44 <i>Sterculia oblongata</i>	
20 <i>Samanea saman</i>	45 <i>Sterculia stipulata</i>	
21 <i>Samanea saman</i> (Yellow Variety)	46 <i>Streblus ilicifolius</i>	
22 <i>Sapium glandulosum</i>	47 <i>Synsepalum dulcificum</i>	
23 <i>Schinus terebinthifolius</i>	48 <i>Syzygium aqueum</i>	
24 <i>Schizolobium parahyba</i>	49 <i>Syzygium borneense</i>	
25 <i>Senna siamea</i>	50 <i>Syzygium jambos</i>	

Trees with dense canopy		
1 <i>Acacia auriculiformis</i>	47 <i>Bauhinia tomentosa</i>	93 <i>Diospyros buxifolia</i>
2 <i>Acacia confusa</i>	48 <i>Blighia sapida</i>	94 <i>Diospyros tristis</i>
3 <i>Adansonia digitata</i>	49 <i>Bouea macrophylla</i>	95 <i>Dipterocarpus baudii</i>
4 <i>Adinandra dumosa</i>	50 <i>Brachychiton rupestris</i>	96 <i>Dipterocarpus oblongifolius</i>
5 <i>Alangium ridleyi</i>	51 <i>Brownea coccinea</i> ssp. <i>capitella</i>	97 <i>Dryobalanops aromatica</i>
6 <i>Aleurites moluccana</i>	52 <i>Brownea grandiceps</i>	98 <i>Duabanga grandiflora</i>
7 <i>Alstonia angustifolia</i>	53 <i>Brownea rosa-de-monte</i>	99 <i>Durio zibethinus</i>
8 <i>Alstonia angustiloba</i>	54 <i>Buchanania arborescens</i>	100 <i>Fagraea crenulata</i>
9 <i>Alstonia scholaris</i>	55 <i>Bucida buceras</i>	101 <i>Fagraea fragrans</i>
10 <i>Alstonia spatulata</i>	56 <i>Butea monosperma</i>	102 <i>Ficus barteri</i>
11 <i>Amesiodendron chinense</i>	57 <i>Caesalpinia ferrea</i>	103 <i>Garcinia atroviridis</i>
12 <i>Amherstia nobilis</i>	58 <i>Caesalpinia sappan</i>	104 <i>Garcinia cymosa</i> forma <i>pendula</i>
13 <i>Anacardium occidentale</i>	59 <i>Callerya atropurpurea</i>	105 <i>Garcinia griffithii</i>
14 <i>Andira inermis</i>	60 <i>Callerya atropurpurea</i> var. <i>xerocarpa</i>	106 <i>Garcinia mangostana</i>
15 <i>Annona cherimola</i>	61 <i>Callistemon citrinus</i>	107 <i>Garcinia subelliptica</i>
16 <i>Annona mucosa</i>	62 <i>Callistemon viminalis</i>	108 <i>Gardenia carinata</i>
17 <i>Antidesma bunius</i>	63 <i>Calophyllum pulcherrimum</i>	109 <i>Gnetum gnemon</i>
18 <i>Antidesma cuspidatum</i>	64 <i>Campnosperma auriculatum</i>	110 <i>Heritiera littoralis</i>
19 <i>Aphanamixis polystachya</i>	65 <i>Capparis micracantha</i>	111 <i>Hevea brasiliensis</i>
20 <i>Aporosa frutescens</i>	66 <i>Carapa guianensis</i>	112 <i>Hopea pubescens</i>
21 <i>Araucaria bidwillii</i>	67 <i>Cassia grandis</i>	113 <i>Hopea sangal</i>
22 <i>Araucaria columnaris</i>	68 <i>Cassia javanica</i> ssp. <i>nodosa</i>	114 <i>Hydnocarpus castanea</i>
23 <i>Araucaria heterophylla</i>	69 <i>Castanopsis inermis</i>	115 <i>Hymenaea courbaril</i>
24 <i>Archidendron clypearia</i>	70 <i>Castanospermum australe</i>	116 <i>Madhuca longifolia</i>
25 <i>Archidendron jiringa</i>	71 <i>Castilla elastica</i>	117 <i>Maranthes corymbosa</i>
26 <i>Archidendron lucidum</i>	72 <i>Casuarina equisetifolia</i>	118 <i>Ochanostachys amentacea</i>
27 <i>Archidendron lucyi</i>	73 <i>Casuarina junghuhniana</i>	119 <i>Palaquium leiocarpum</i>
28 <i>Arfeuillea arborescens</i>	74 <i>Casuarina sumatrana</i>	120 <i>Palaquium oxleyanum</i>
29 <i>Arthrophyllum diversifolium</i>	75 <i>Cecropia peltata</i>	121 <i>Shorea leprosula</i>
30 <i>Artocarpus altilis</i>	76 <i>Cerbera manghas</i>	122 <i>Sterculia parvifolia</i>
31 <i>Artocarpus heterophyllus</i>	77 <i>Cerbera odollam</i>	123 <i>Sterculia rubiginosa</i>
32 <i>Asteromyrtus symphyocarpa</i>	78 <i>Chrysophyllum cainito</i>	124 <i>Streblus elongatus</i>
33 <i>Averrhoa bilimbi</i>	79 <i>Chukrasia tabularis</i>	125 <i>Strombosia javanica</i>
34 <i>Averrhoa carambola</i>	80 <i>Cleistanthus sumatranus</i>	126 <i>Syzygium aromaticum</i>
35 <i>Azadirachta excelsa</i>	81 <i>Clerodendrum laevifolium</i>	127 <i>Syzygium campanulatum</i>
36 <i>Azadirachta indica</i>	82 <i>Clitoria fairchildiana</i>	128 <i>Syzygium campanulatum</i> (Dark Red)
37 <i>Baccaurea motleyana</i>	83 <i>Clusia rosea</i>	129 <i>Syzygium cumini</i>
38 <i>Baccaurea parviflora</i>	84 <i>Clusia rosea</i> 'Aureo-Marginata'	130 <i>Syzygium glaucum</i>
39 <i>Baekkea frutescens</i>	85 <i>Cola nitida</i>	131 <i>Syzygium polyanthum</i>
40 <i>Baikiaea insignis</i>	86 <i>Conocarpus erectus</i> var. <i>sericeus</i>	132 <i>Syzygium zeylanicum</i>
41 <i>Barringtonia acutangula</i>	87 <i>Copaifera officinalis</i>	133 <i>Tabebuia pallida</i>
42 <i>Barringtonia asiatica</i>	88 <i>Cordia subcordata</i>	134 <i>Tabebuia riparia</i>
43 <i>Barringtonia calyptrata</i>	89 <i>Corymbia ptychocarpa</i>	135 <i>Tabebuia rosea</i>
44 <i>Barringtonia edulis</i>	90 <i>Couroupita guianensis</i>	
45 <i>Barringtonia racemosa</i>	91 <i>Cryptomeria japonica</i>	
46 <i>Bauhinia monandra</i>	92 <i>Dillenia indica</i>	

Trees with intermediate canopy		
1 <i>Agathis borneensis</i>	47 <i>Dalbergia latifolia</i>	93 <i>Ficus auriculata</i>
2 <i>Aglaia formosana</i>	48 <i>Dalbergia oliveri</i>	94 <i>Ficus benghalensis</i>
3 <i>Anaxagorea javanica</i>	49 <i>Delonix regia</i>	95 <i>Ficus benghalensis</i>
4 <i>Annona squamosa</i>	50 <i>Dillenia alata</i>	96 <i>Ficus benghalensis</i> 'Krishnae'
5 <i>Aquilaria malaccensis</i>	51 <i>Dillenia grandifolia</i>	97 <i>Ficus benamina</i>
6 <i>Ardisia elliptica</i>	52 <i>Dillenia ingens</i>	98 <i>Ficus benamina</i> 'Variegated White'
7 <i>Ardisia lanceolata</i>	53 <i>Dillenia ovata</i>	99 <i>Ficus celebensis</i>
8 <i>Bauhinia purpurea</i>	54 <i>Dillenia philippinensis</i>	100 <i>Ficus destruens</i>
9 <i>Bauhinia purpurea</i> x <i>variegata</i> 'Blakeana'	55 <i>Dillenia reticulata</i>	101 <i>Ficus elastica</i>
10 <i>Brachychiton acerifolius</i>	56 <i>Dillenia suffruticosa</i>	102 <i>Ficus elastica</i> 'Black Burgundy'
11 <i>Bridelia tomentosa</i>	57 <i>Dimocarpus longan</i>	103 <i>Ficus elastica</i> cultivar
12 <i>Bucida molinetii</i>	58 <i>Diospyros blancoi</i>	104 <i>Ficus grossularioides</i>
13 <i>Bunchosia armeniaca</i>	59 <i>Diospyros diepenhorstii</i>	105 <i>Ficus kurzii</i>
14 <i>Caesalpinia coriaria</i>	60 <i>Diospyros ferrea</i>	106 <i>Ficus microcarpa</i>
15 <i>Calliandra tergemina</i> var. <i>emarginata</i>	61 <i>Diospyros malabarica</i>	107 <i>Ficus microcarpa</i> 'Golden'
16 <i>Calophyllum inophyllum</i>	62 <i>Dipterocarpus grandiflorus</i>	108 <i>Ficus religiosa</i>
17 <i>Calophyllum soulattri</i>	63 <i>Dolichandrone spathacea</i>	109 <i>Ficus sabre</i>
18 <i>Calyculophyllum spruceanum</i>	64 <i>Dovyalis hebecarpa</i>	110 <i>Ficus superba</i>
19 <i>Cananga odorata</i>	65 <i>Dracaena maingayi</i>	111 <i>Ficus variegata</i>
20 <i>Carallia brachiata</i>	66 <i>Dryobalanops oblongifolia</i>	112 <i>Filicium decipiens</i>
21 <i>Carallia brachiata</i> 'Honiara'	67 <i>Dyera costulata</i>	113 <i>Firmiana malayana</i>
22 <i>Carallia suffruticosa</i>	68 <i>Dysoxylum cauliflorum</i>	114 <i>Flacourtia inermis</i>
23 <i>Carica papaya</i>	69 <i>Ebenopsis ebano</i>	115 <i>Flacourtia rukam</i>
24 <i>Cassia fistula</i>	70 <i>Elaeocarpus angustifolius</i>	116 <i>Garcinia dulcis</i>
25 <i>Ceiba pentandra</i>	71 <i>Elaeocarpus grandiflorus</i>	117 <i>Garcinia nitida</i>
26 <i>Champerea manillana</i>	72 <i>Elaeocarpus hainanensis</i>	118 <i>Garcinia parvifolia</i>
27 <i>Chorisia speciosa</i>	73 <i>Elaeocarpus mastersii</i>	119 <i>Garcinia xanthochymus</i>
28 <i>Cinnamomum camphora</i>	74 <i>Elaeocarpus petiolatus</i>	120 <i>Gardenia tubifera</i>
29 <i>Cinnamomum iners</i>	75 <i>Elaeocarpus stipularis</i>	121 <i>Gironniera nervosa</i>
30 <i>Cinnamomum verum</i>	76 <i>Elateriospermum tapos</i>	122 <i>Gliricidia sepium</i>
31 <i>Citharexylum spinosum</i>	77 <i>Enterolobium cyclocarpum</i>	123 <i>Glochidion littorale</i>
32 <i>Claoxylon indicum</i>	78 <i>Eriobotrya japonica</i>	124 <i>Glochidion superbum</i>
33 <i>Coccoloba uvifera</i>	79 <i>Erythrina fusca</i>	125 <i>Glycosmis perakensis</i>
34 <i>Cochlospermum religiosum</i>	80 <i>Erythrina variegata</i>	126 <i>Gmelina arborea</i>
35 <i>Coffea canephora</i>	81 <i>Erythrina vespertilio</i>	127 <i>Gmelina dalrympleana</i>
36 <i>Coffea liberica</i>	82 <i>Erythrophloeum guineense</i>	128 <i>Gmelina elliptica</i>
37 <i>Cola gigantea</i>	83 <i>Erythroxylum cuneatum</i>	129 <i>Grevillea baileyana</i>
38 <i>Cordia sebestena</i>	84 <i>Eucalyptus alba</i>	130 <i>Grevillea robusta</i>
39 <i>Cordia sebestena</i> 'Aurea'	85 <i>Euclina longiflora</i>	131 <i>Guaiacum officinale</i>
40 <i>Crateva religiosa</i>	86 <i>Eugenia dombeyi</i>	132 <i>Guettarda speciosa</i>
41 <i>Cratoxylum cochinchinense</i>	87 <i>Eurycoma longifolia</i>	133 <i>Gustavia superba</i>
42 <i>Cratoxylum formosum</i>	88 <i>Eusideroxylum zwageri</i>	134 <i>Gymnostoma nobile</i>
43 <i>Crescentia cujete</i>	89 <i>Evodiella muelleri</i>	135 <i>Gymnostoma rumphianum</i>
44 <i>Cynometra cauliflora</i>	90 <i>Excoecaria agallocha</i>	136 <i>Gymnostoma sumatrana</i>
45 <i>Cynometra ramiflora</i>	91 <i>Fagraea auriculata</i>	137 <i>Gynotroches axillaris</i>
46 <i>Dalbergia cochinchinensis</i>	92 <i>Fagraea racemosa</i>	138 <i>Heritiera elata</i>

Trees with intermediate canopy		
139 <i>Hibiscus tiliaceus</i>	185 <i>Mangifera caesia</i>	231 <i>Plumeria rubra</i>
140 <i>Hibiscus tiliaceus</i> 'Tricolor'	186 <i>Mangifera foetida</i>	232 <i>Podocarpus macrophyllus</i>
141 <i>Hopea helferi</i>	187 <i>Mangifera indica</i>	233 <i>Podocarpus neriifolius</i>
142 <i>Hopea mengarawan</i>	188 <i>Manilkara zapota</i>	234 <i>Podocarpus polystachyus</i>
143 <i>Hopea nutans</i>	189 <i>Maniltoa browneoides</i>	235 <i>Podocarpus rumphii</i>
144 <i>Hopea odorata</i>	190 <i>Maniltoa lenticellata</i>	236 <i>Polyalthia longifolia</i>
145 <i>Hopea wightiana</i>	191 <i>Melaleuca cajuputi</i>	237 <i>Polyalthia longifolia</i> var. <i>pendula</i>
146 <i>Horsfieldia crassifolia</i>	192 <i>Melaleuca dealbata</i>	238 <i>Polyalthia rumphii</i>
147 <i>Horsfieldia irya</i>	193 <i>Melaleuca styphelioides</i>	239 <i>Polyalthia sclerophylla</i>
148 <i>Horsfieldia sucosa</i>	194 <i>Memecylon caeruleum</i>	240 <i>Polyalthia stenopetala</i>
149 <i>Hura crepitans</i>	195 <i>Michelia champaca</i>	241 <i>Pometia pinnata</i>
150 <i>Ilex cymosa</i>	196 <i>Millettia pinnata</i>	242 <i>Pouteria cainito</i>
151 <i>Intsia bijuga</i>	197 <i>Millingtonia hortensis</i>	243 <i>Pouteria campechiana</i>
152 <i>Intsia palembanica</i>	198 <i>Mimusops elengi</i>	244 <i>Pouteria gardneriana</i>
153 <i>Ixonanthes icosandra</i>	199 <i>Monocarpia marginalis</i>	245 <i>Pouteria obovata</i>
154 <i>Jacaranda obtusifolia</i>	200 <i>Morella esculenta</i>	246 <i>Pterocarpus indicus</i>
155 <i>Jatropha gossypifolia</i>	201 <i>Morinda citrifolia</i>	247 <i>Pterocarpus indicus</i> var. <i>pendula</i>
156 <i>Kailarsenia</i> sp.	202 <i>Morus alba</i>	248 <i>Pterospermum acerifolium</i>
157 <i>Khaya grandifoliola</i>	203 <i>Muntingia calabura</i>	249 <i>Pterygota alata</i>
158 <i>Khaya senegalensis</i>	204 <i>Myristica fragrans</i>	250 <i>Randia anisophylla</i>
159 <i>Kigelia africana</i>	205 <i>Nauclea orientalis</i>	251 <i>Reevesia thyrsoidea</i>
160 <i>Kleinhovia hospita</i>	206 <i>Neisosperma oppositifolium</i>	252 <i>Salix babylonica</i>
161 <i>Knema globularia</i>	207 <i>Neobalanocarpus heimii</i>	253 <i>Sandoricum koetjape</i>
162 <i>Koompassia excelsa</i>	208 <i>Neolamarckia cadamba</i>	254 <i>Saraca cauliflora</i>
163 <i>Koompassia malaccensis</i>	209 <i>Nephelium lappaceum</i>	255 <i>Saraca declinata</i>
164 <i>Kopsia flavida</i>	210 <i>Nephelium ramboutan-ake</i>	256 <i>Saraca indica</i>
165 <i>Kopsia fruticosa</i>	211 <i>Pachira aquatica</i>	257 <i>Scaphium macropodum</i>
166 <i>Kopsia singaporensis</i>	212 <i>Pachira insignis</i>	258 <i>Schefflera actinophylla</i>
167 <i>Lagerstroemia floribunda</i>	213 <i>Palaquium obovatum</i>	259 <i>Sonneratia caseolaris</i>
168 <i>Lagerstroemia langkawiensis</i>	214 <i>Pangium edule</i>	260 <i>Spathodea campanulata</i>
169 <i>Lagerstroemia loudonii</i>	215 <i>Paraserianthes falcataria</i>	261 <i>Stenocarpus sinuatus</i>
170 <i>Lagerstroemia speciosa</i>	216 <i>Parkia speciosa</i>	262 <i>Stereospermum fimbriatum</i>
171 <i>Lagerstroemia speciosa</i>	217 <i>Parkia timoriana</i>	263 <i>Suregada multiflora</i>
172 <i>Lagerstroemia tomentosa</i>	218 <i>Peltophorum pterocarpum</i>	264 <i>Swietenia macrophylla</i>
173 <i>Lansium domesticum</i>	219 <i>Pentace curtisii</i>	265 <i>Swietenia mahogani</i>
174 <i>Lecythis ollaria</i>	220 <i>Pentace triptera</i>	266 <i>Syzygium grande</i>
175 <i>Lepisanthes alata</i>	221 <i>Phaleria clerodendron</i>	267 <i>Syzygium lineatum</i>
176 <i>Leptospermum brachyandrum</i>	222 <i>Phyllanthus pectinatus</i>	268 <i>Syzygium malaccense</i>
177 <i>Lithocarpus elegans</i>	223 <i>Pisonia grandis</i>	269 <i>Terminalia calamansanai</i>
178 <i>Lophanthera lactescens</i>	224 <i>Pithecellobium dulce</i>	270 <i>Terminalia phellocarpa</i>
179 <i>Macadamia integrifolia</i>	225 <i>Pittosporum ferrugineum</i>	271 <i>Tristaniaopsis laurina</i>
180 <i>Macaranga bancana</i>	226 <i>Pittosporum tobira</i>	272 <i>Tristaniaopsis whiteana</i>
181 <i>Magnolia coco</i>	227 <i>Platycladus orientalis</i>	273 <i>Xanthophyllum obscurum</i>
182 <i>Magnolia figo</i>	228 <i>Pleiocarpa mutica</i>	274 <i>Xanthostemon chrysanthus</i>
183 <i>Magnolia grandiflora</i>	229 <i>Ploiarium alternifolium</i>	275 <i>Xanthostemon verticillatus</i>
184 <i>Majidea zanguebarica</i>	230 <i>Plumeria obtusa</i>	

Appendix 2 List Of Palms (Solitary or Cluster)

Examples of Solitary Palms	
1 <i>Archontophoenix alexandrae</i>	13 <i>Livistona chinensis</i>
2 <i>Bentinckia nicobarica</i>	14 <i>Livistona rotundifolia</i>
3 <i>Bismarckia nobilis</i> 'Silver'	15 <i>Normanbya normanbyi</i>
4 <i>Carpentaria acuminata</i>	16 <i>Phoenicophorium borsigianum</i>
5 <i>Cocos nucifera</i>	17 <i>Phoenix roebelenii</i>
6 <i>Cocos nucifera</i> 'Thai Fragrance'	18 <i>Phoenix sylvestris</i>
7 <i>Dypsis decaryi</i>	19 <i>Pritchardia pacifica</i>
8 <i>Dypsis leptocheilos</i>	20 <i>Roystonea oleracea</i>
9 <i>Dypsis madagascariensis</i>	21 <i>Thrinax parviflora</i>
10 <i>Elaeis guineensis</i>	22 <i>Veitchia merrillii</i>
11 <i>Latania verschoffeltii</i>	23 <i>Wodyetia bifurcata</i>
12 <i>Licuala grandis</i>	

Examples of Cluster Palms	
1 <i>Areca vestiaria</i>	10 <i>Phoenix reclinata</i>
2 <i>Arenga hookeriana</i>	11 <i>Ptychosperma macarthurii</i>
3 <i>Bactris guineensis</i>	12 <i>Raphis excelsa</i>
4 <i>Caryota mitis</i>	13 <i>Raphis humilis</i>
5 <i>Dypsis lutescens</i>	14 <i>Raphis multifida</i>
6 <i>Euterpe edulis</i>	15 <i>Rhopaloblaste singaporensis</i>
7 <i>Licuala spinosa</i>	16 <i>Salacca magnifica</i>
8 <i>Nypa fruticans</i>	17 <i>Serenoa repens</i>
9 <i>Oncosperma tigillarum</i>	

Appendix 3. List Of Shrubs (Monocotyledons or Dicotyledons)

To identify the LAI value of shrubs not listed in the table, the botanical class of dicot or monocot can be determined from the Plants Database accessible from this website (<http://plants.usda.gov>).

Monocotyledon Shrubs

Families of the Monocotyledonae include Alismataceae, Amaryllidaceae, Araceae, Arecaceae, Bromeliaceae, Commelinaceae, Cyperaceae, Dioscoreaceae, Hydrocharitaceae, Iridaceae, Lemnaceae, Liliaceae, Musaceae, Orchidaceae, Pandanaceae, Pontederiaceae, Poaceae, Typhaceae

Examples of Monocotyledons Shrubs	
1 <i>Aglaonema commutatum</i>	37 <i>Dieffenbachia cultivar</i>
2 <i>Aglaonema costatum</i> forma <i>foxii</i>	38 <i>Dieffenbachia</i> 'Mosaic'
3 <i>Aglaonema</i> 'Donna Carmen'	39 <i>Dieffenbachia picta</i> 'Roehrsii'
4 <i>Aglaonema</i> 'Earnesto Hybrid'	40 <i>Dieffenbachia</i> 'Silver'
5 <i>Aglaonema hybrid</i>	41 <i>Dieffenbachia</i> 'Sublime'
6 <i>Aglaonema nitidum</i>	42 <i>Dieffenbachia</i> x <i>bausei</i>
7 <i>Aglaonema rex</i>	43 <i>Epidendrum ibaguense</i>
8 <i>Aglaonema siamense</i>	44 <i>Epipremnum aureum</i>
9 <i>Alocasia cucullata</i>	45 <i>Epipremnum pinnatum</i>
10 <i>Alocasia denudata</i>	46 <i>Grammatophyllum speciosum</i>
11 <i>Alocasia</i> 'Hana'	47 <i>Homalomena rubescens</i>
12 <i>Alocasia</i> 'Hilo Beauty'	48 <i>Homalomena</i> 'Selby'
13 <i>Alocasia macrorrhizos</i>	49 <i>Homalomena wallichii</i>
14 <i>Alocasia</i> x <i>amazonica</i>	50 <i>Homalomena</i> 'Emerald Gem'
15 <i>Amorphophallus paeoniifolius</i>	51 <i>Ludisia discolor</i>
16 <i>Amorphophallus prainii</i>	52 <i>Ludisia discolor</i>
17 <i>Anthurium andraeanum</i>	53 <i>Monstera deliciosa</i>
18 <i>Anthurium andraeanum</i> 'Album'	54 <i>Montrichardia arborescens</i>
19 <i>Anthurium andraeanum</i> 'Purple'	55 <i>Philodendron erubescens</i>
20 <i>Anthurium crystallinum</i>	56 <i>Philodendron mamei</i>
21 <i>Anthurium cultivars</i>	57 <i>Philodendron xanadu</i>
22 <i>Anthurium plowmanii</i>	58 <i>Pistia stratiotes</i>
23 <i>Anubias barteri</i> var. <i>glabra</i>	59 <i>Schismatoglottis wallichii</i>
24 <i>Arundina graminifolia</i>	60 <i>Scindapsus pictus</i> 'Argyraeus'
25 <i>Caladium bicolor</i>	61 <i>Spathiphyllum cannifolium</i>
26 <i>Caladium humboldtii</i>	62 <i>Spathiphyllum wallisii</i>
27 <i>Caladium lindenii</i>	63 <i>Spathoglottis gracilis</i>
28 <i>Colocasia esculenta</i>	64 <i>Spathoglottis kimballiana</i>
29 <i>Colocasia esculenta</i> 'Blackmagic'	65 <i>Spathoglottis plicata</i>
30 <i>Dendrobium crumenatum</i>	66 <i>Spathoglottis unguiculata</i>
31 <i>Dieffenbachia amoena</i>	67 <i>Syngonium cultivar</i>
32 <i>Dieffenbachia amoena</i> 'Tropic Snow'	68 <i>Vanda</i> 'Miss Joaquim'
33 <i>Dieffenbachia</i> 'Besar Putih'	69 <i>Vanilla griffithii</i>
34 <i>Dieffenbachia</i> 'Camilla'	70 <i>Vanilla planifolia</i>
35 <i>Dieffenbachia candida</i>	71 <i>Zamioculcas zamiifolia</i>
36 <i>Dieffenbachia</i> 'Cheetah'	72 <i>Zephyranthes rosea</i>

Dicotyledon Shrubs

Families of the Dicotyledonae include Anacardiaceae, Asteraceae, Brassicaceae, Cactaceae, Cucurbitaceae, Euphorbiaceae, Fabaceae, Fagaceae, Linaceae, Malvaceae, Oleaceae, Rosaceae, Rubiaceae, Rutaceae, Solanaceae, Theaceae, Vitaceae

Examples of Dicotyledon Shrubs	
1 <i>Acalypha chazmaedrifolia</i>	32 <i>Ixora hybrid</i>
2 <i>Acalypha godseffiana</i> 'Heterophylla'	33 <i>Ixora javanica</i>
3 <i>Acalypha hispida</i>	34 <i>Ixora odorata</i>
4 <i>Acalypha siamensis</i>	35 <i>Ixora pendula</i>
5 <i>Acalypha wilkesiana</i>	36 <i>Ixora siamensis</i>
6 <i>Agrostistachys longifolia</i>	37 <i>Jatropha gossypifolia</i>
7 <i>Borreria laevicaulis</i>	38 <i>Jatropha integerrima</i>
8 <i>Breynia disticha</i>	39 <i>Jatropha integerrima</i> 'Pink'
9 <i>Bridelia tomentosa</i>	40 <i>Jatropha podagrica</i>
10 <i>Carphalea kirondron</i>	41 <i>Lagenaria siceraria</i>
11 <i>Codiaeum variegatum</i>	42 <i>Manettia luteorubra</i>
12 <i>Coffea canephora</i>	43 <i>Manihot esculenta</i>
13 <i>Coffea liberica</i>	44 <i>Manihot glaziovii</i>
14 <i>Croton tiglium</i>	45 <i>Momordica charantia</i>
15 <i>Euclinia longiflora</i>	46 <i>Momordica cochinchinensis</i>
16 <i>Euphorbia antiquorum</i>	47 <i>Morinda citrifolia</i>
17 <i>Euphorbia cotinifolia</i>	48 <i>Morinda umbellata</i>
18 <i>Euphorbia hirta</i>	49 <i>Mussaenda</i> 'Calcutta'
19 <i>Euphorbia lactea</i>	50 <i>Mussaenda philippica</i> 'Aurorae'
20 <i>Euphorbia milii</i>	51 <i>Myrmecodia tuberosa</i>
21 <i>Euphorbia neriifolia</i>	52 <i>Pedilanthus tithymaloides</i>
22 <i>Euphorbia pulcherrima</i>	53 <i>Pentas lanceolata</i>
23 <i>Euphorbia tirucalli</i>	54 <i>Phyllanthus amarus</i>
24 <i>Excoecaria cochinchinensis</i>	55 <i>Phyllanthus cochinchinensis</i>
25 <i>Gardenia augusta</i>	56 <i>Phyllanthus cuscuteaeflorus</i>
26 <i>Gardenia scabrella</i>	57 <i>Phyllanthus myrtifolius</i>
27 <i>Gardenia tubifera</i>	58 <i>Phyllanthus pectinatus</i>
28 <i>Hamelia patens</i>	59 <i>Phyllanthus pulcher</i>
29 <i>Hoffmannia bullata</i>	60 <i>Phyllanthus urinaria</i>
30 <i>Ixora caesia</i>	61 <i>Ricinus communis</i>
31 <i>Ixora grandifolia</i>	62 <i>Rothmannia macrophylla</i>